Such was the case until one fateful night in December 1979, when Alan Guth, a physics postdoctoral fellow working at the Stanford Linear Accelerator Center (he is now a professor at MIT), showed that we can do better. Much better. Although there are details that today, more than two decades later, have yet to be resolved fully, Guth made a discovery that finally filled the cosmological silence by providing the big bang with a bang, and one that was bigger than anyone expected.

Guth was not trained as a cosmologist. His specialty was particle physics, and in the late 1970s, together with Henry Tye from Cornell University, he was studying various aspects of Higgs fields in grand unified theories. Remember from the last chapter's discussion of spontaneous symmetry breaking that a Higgs field contributes the least possible energy it can to a region of space when its value settles down to a particular nonzero number (a number that depends on the detailed shape of its potential energy bowl). In the early universe, when the temperature was extraordinarily high, we discussed how the value of a Higgs field would wildly fluctuate from one number to another, like the frog in the hot metal bowl whose legs were being singed, but as the universe cooled, the Higgs would roll down the bowl to a value that would minimize its energy.

Guth and Tye studied reasons why the Higgs field might be delayed in reaching the least energetic configuration (the bowl's valley in Figure 9.1c). If we apply the frog analogy to the question Guth and Tye asked, it was this: what if the frog, in one of its earlier jumps when the bowl was starting to cool, just happened to land on the central plateau? And what if, as the bowl continued to cool, the frog hung out on the central plateau (leisurely eating worms), rather than sliding down to the bowl's valley? Or, in physics terms, what if a fluctuating Higgs field's value should land on the energy bowl's central plateau and remain there as the universe continues to cool? If this happens, physicists say that the Higgs field has *supercooled*, indicating that even though the temperature of the universe has dropped to the point where you'd expect the Higgs value to approach the low-energy valley, it remains trapped in a higher-energy configuration. (This is analogous to highly purified water, which can be supercooled below 0 degrees Celsius, the temperature at which you'd expect it to turn into ice, and yet remain liquid because the formation of ice requires small impurities around which the crystals can grow.)